Project Management System (PMS)

1. What is a Project Management System?

A **Project Management System (PMS)** is a software tool that helps a company plan, organize, and manage its projects.

In simple words, it is a system that helps teams:

- Know what work needs to be done
- Know who is responsible for each task
- Track deadlines
- Monitor progress

It keeps all project-related information in one place.

2. Why Do We Need a Project Management System?

Without a PMS:

- Work gets confused
- People forget deadlines
- Tasks are not clearly assigned
- Managers cannot track progress

With a PMS:

- Work is organized
- Every task has a responsible person
- Deadlines are visible
- Progress can be tracked easily

It improves teamwork and productivity.

3. How Does a Project Management System Work?

A PMS works in a structured way.

Step 1: Users in the System

There are different types of users:

- **Admin** – Controls the system
- **Team Lead** – Manages team tasks
- **Employee** – Completes assigned tasks

Each person has a specific role and responsibility.

Step 2: Project Creation

The Admin creates a project.

Example:
Project Name: Website Development
Description: Build a company website

A project is the main goal.

Step 3: Task Creation

The Admin creates tasks under the project and assigns them to a Team Lead.

Example:
Task: Develop Login System
Assigned to: Team Lead

Step 4: Subtask Assignment

The Team Lead divides the main task into smaller subtasks and assigns them to employees.

Example:-
Subtask 1: Design Login Page
Subtask 2: Create Login API

Each subtask has:

- Deadline
- Status (Pending / In Progress / Completed)

Step 5: Tracking Progress

Employees update their task status.

Team Lead checks subtask progress.

Admin monitors overall project progress.

This way, everyone knows:

- What work is done
- What work is pending
- Who is responsible

4. Real-Life Example

Imagine a company building a mobile app.

Admin:

- Creates Project: "Mobile App Development"
- Assigns task: "Payment Module" to Team Lead

Team Lead:

- Creates Subtask: "Payment UI" → Assigned to Employee A
- Creates Subtask: "Payment API" → Assigned to Employee B

Employees complete their tasks and update status.

Admin can see the full progress of the project.

5. Benefits of a Project Management System

- Better organization
- Clear responsibilities
- Improved teamwork
- Easy tracking of deadlines
- Higher productivity

6. Examples of Project Management Systems

Some popular project management tools are:

- Jira
- Asana
- Trello
- Monday.com

These tools help companies manage projects in a structured way.

❖ Role-Based Sections in Project Management System

The system has three main roles:

- Admin
- Team Lead
- Employee

Each role has a different view and different permissions.

1. Admin Mode

What Admin Sees (Dashboard View)

When Admin logs in, they can see:

- Total number of projects
- Total number of Team Leads
- Total number of Employees
- List of all projects
- Project status (Active / Completed)
- Tasks overview
- Overdue tasks
- Recently created tasks

Admin can also see:

- Which project is assigned to which Team Lead
- Which Team Lead is handling how many tasks
- Overall project progress

What Admin Can Do

Admin has full control over the system.

Admin can:

✔Create, update, delete projects

✔Create main tasks

✔Assign tasks to Team Leads

✔Add new Team Leads and Employees

✔Edit user details

✔Remove users (if needed)

✔View full system reports

✔Monitor project progress

Admin cannot:

- Directly interfere in employee subtasks (optional depending on company policy)

2. <u>Team Lead Mode</u>

What Team Lead Sees

When Team Lead logs in, they see:

- Projects assigned to them
- Main tasks assigned by Admin
- Task deadlines
- Task status
- List of employees under them
- Subtask progress

They cannot see:

- All system users
- Projects not assigned to them

What Team Lead Can Do

Team Lead can:

✔View assigned projects

✔View assigned main tasks

✔Create subtasks

✔Assign subtasks to employees

✔Update main task status

✔Monitor employee progress

✔Change subtask deadlines (if needed)

Team Lead cannot:

- Create new projects

- Add new users
- Assign tasks to other Team Leads

3. <u>Employee Mode</u>

What Employee Sees

When Employee logs in, they see:

- List of subtasks assigned to them
- Deadline for each subtask
- Status of their tasks
- Project name related to task

They cannot see:

- Other employees' tasks
- Full project details
- System-wide data

What Employee Can Do

Employee can:

✔View assigned subtasks

✔Update task status (Pending / In Progress / Completed)

✔Add comments or updates

✔Upload files (optional feature)

Employee cannot:

- Create tasks
- Assign tasks
- View other employees’ work
- Create projects

Project Manager Role (Optional)

If needed, a Project Manager can:

- Manage multiple Team Leads
- View performance reports
- Approve project completion

This role sits between Admin and Team Lead.

❖ How the Window/Dashboard Should Feel

Admin Dashboard → System overview (big picture)
Team Lead Dashboard → Task management view
Employee Dashboard → Work assignment view

Each role sees only what is necessary.
This is called **Role-Based Access Control (RBAC)**.